Recall

If x is T-periodic and $\omega = \frac{2\pi}{T}$, then:

the n^{th} Fourier series coefficient of x is

$$c_n = \frac{1}{T}\langle x, e^{jn\omega t}\rangle, \quad n \in \mathbb{Z}$$

$$= \frac{1}{T}\int_0^T x(t)\underbrace{e^{-jn\omega t}}\,dt.$$

Fund period $= \frac{T}{|n|} = \frac{2\pi}{|n|\omega}$

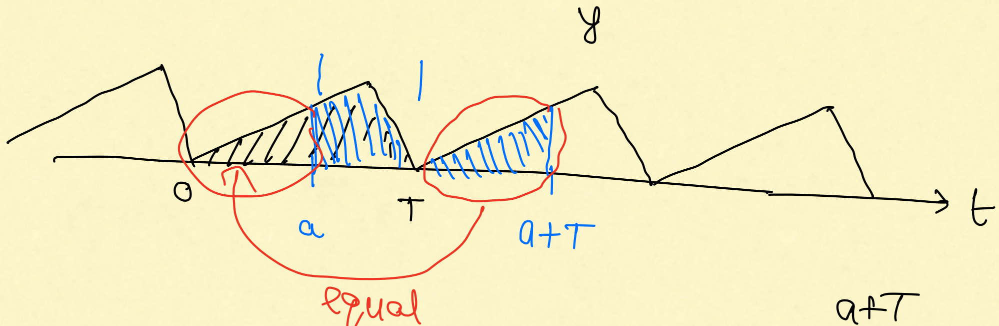
$\leq$

a Period $= T$

Remarks:

① If y is T-periodic

say we want $\int_0^T y(t)\,dt$.



$\Rightarrow$ for any real a, $\int_0^T y(t)\,dt = \int_a^{a+T} y(t)\,dt$.

$$\Rightarrow c_n = \frac{1}{T}\int_a^{a+T} x(t)e^{-jn\omega t}\,dt$$

Typically, $a = 0$ or $a = -T/2$

② We say that a signal y is even if

$$y(-t) = y(t)$$

and we say that y is odd if

$$y(-t) = -y(t)$$

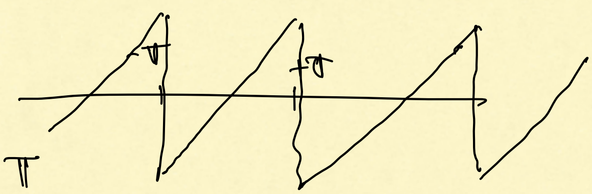
Even: $$\int_{-T/2}^{T/2} y(t)\,dt = 2\int_{0}^{T/2} y(t)\,dt$$

Odd: $$\int_{-T/2}^{+T/2} y(t)\,dt = 0$$

(3) $$\int u\,dv = uv - \int v\,du.$$

(I) Examples

(1) $T = 2\pi \Rightarrow \omega = \frac{2\pi}{T} = 1$



$x(t) = t$ for $-\pi \le t < \pi$

This is called the sawtooth waveform

n=0: $$c_0 = \frac{1}{2\pi}\int_{-\pi}^{+\pi} t\cdot e^{-j0t}\,dt = \frac{1}{2\pi}\int_{-\pi}^{+\pi} t\,dt = 0$$

n≠0: $$c_n = \frac{1}{2\pi}\int_{-\pi}^{+\pi} t\cdot e^{-jnt}\,dt.$$

$$\left\{ \int e^{zt}\,dt = \frac{e^{zt}}{z} \right.$$ for any $z \neq 0$.

$$= \frac{1}{2\pi}\int_{-\pi}^{+\pi} \underbrace{t}_{u}\cdot d\underbrace{\left[\frac{e^{-jnt}}{-jn}\right]}_{v}.$$

$$2\pi c_n = \left[\frac{t\cdot e^{-jnt}}{-jn}\right]_{-\pi}^{+\pi} + \frac{1}{jn}\cancelto{0}{\int_{-\pi}^{+\pi} e^{-jnt}\,dt}.$$

$$= \frac{j}{n}\left[\pi\cdot e^{-jn\pi} + \pi\cdot e^{+jn\pi}\right].$$

$$= \frac{j\pi}{n} 2\cos(n\pi) = \frac{j\pi}{n} \cdot 2 \cdot (-1)^n$$

$$\Rightarrow \quad c_n = \frac{j \cdot (-1)^n}{n}.$$

$$\therefore \quad c_n = \begin{cases} 0, & \text{if } n = 0 \\ \frac{j(-1)^n}{n}, & \text{if } n \neq 0. \end{cases}$$

From Parsevals: $\|x\|^2 = T \sum_n |c_n|^2$

$$LHS = \int_{-\pi}^{+\pi} t^2 dt = 2\pi^3/3.$$

$$RHS = 2\pi \cdot 2 \times \sum_{n=1}^{\infty} \frac{1}{n^2}$$

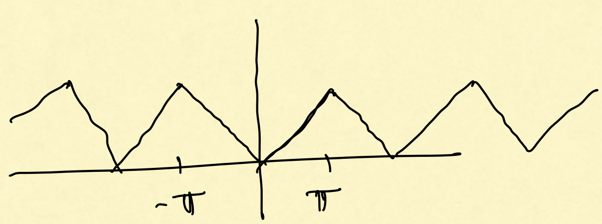
$$\cancel{4\pi} \sum_{n \geq 1} \frac{1}{n^2} = \frac{\cancel{2\pi^3}\, \pi^2}{3 \times 2}$$

$$\therefore \sum_{n \geq 1} \frac{1}{n^2} = \pi^2/6.$$

$$\boxed{\sum_{n=1} \frac{1}{n^4} = ?}$$

② Triangular Wave forms

$T = 2\pi, \quad \omega = 1$



$x(t) = |t|, \quad -\pi \leq t < \pi$

$\underline{n=0}$:

$$C_0 = \frac{1}{2\pi}\int_{-\pi}^{+\pi} |t|\, dt$$

$$= \frac{1}{\pi}\int_{0}^{\pi} t\, dt$$

$$= \frac{1}{\pi} \times \pi^2/2 = \pi/2 .$$

$\underline{n \neq 0}$:

$$2\pi C_n = \int_{-\pi}^{+\pi} |t| \cdot e^{-jn\omega t}\, dt$$

$$= \int_{-\pi}^{+\pi} |t| \cdot \left[\cos nt - j \sin nt\right] dt$$

$$= \int_{-\pi}^{+\pi} |t| \cos(nt)\, dt - j \underbrace{\int_{-\pi}^{+\pi} |t| \cdot \cancel{\sin(nt)}\, dt}_{0}$$

$$2\pi C_n = 2\int_{0}^{\pi} t \cos nt\, dt$$

$$\pi C_n = \int_{0}^{\pi} t \cdot \cos(nt)\, dt .$$

$$= \int_{0}^{\pi} \underbrace{t}_{u} \cdot d\Big[\underbrace{\frac{\sin(nt)}{n}}_{v}\Big]$$

$$= \underbrace{\left[t \cdot \cancel{\frac{\sin nt}{n}}\right]_0^{\pi}}_{0} - \int_{0}^{\pi} \frac{\sin nt}{n}\, dt .$$

$$= \frac{1}{n^2}\left[\cos nt\right]_0^{\pi} = \frac{1}{n^2}\left[(-1)^n - 1\right]$$

$$\therefore \quad C_n = \begin{cases} \pi/2 , & n = 0 \\ 0 , & n \neq 0, \text{ even} \\ \dfrac{-2}{\pi n^2} , & n \neq 0, \text{ odd} \end{cases}$$

II. Basic Properties. (Sec 3.5 of OWN)

Notation: If c_n, $n \in \mathbb{Z}$, are the Fourier series coeff. of x we denote this

$$x \xleftrightarrow{FS} c_n$$

① Linearity

If x and y are T-periodic.

$$x \xleftrightarrow{FS} c_n$$

$$y \xleftrightarrow{FS} d_n$$

$$\alpha x + \beta y \xleftrightarrow{FS} ? \quad , \quad \alpha, \beta \in \mathbb{C}$$

n^{th} FS coeff. for $\alpha x + \beta y$ is

$$\frac{1}{T}\langle \alpha x + \beta y, e^{jn\omega t} \rangle$$

$$= \alpha \frac{1}{T}\langle x, e^{jn\omega t}\rangle + \beta \frac{1}{T}\langle y, e^{jn\omega t}\rangle$$

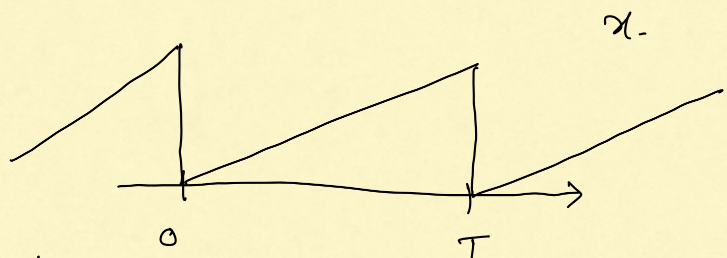
$$= \alpha c_n + \beta d_n.$$

② Time-shifting

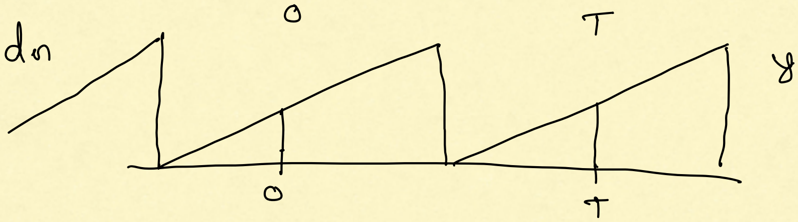
$$y(t) = x(t - t_0)$$

$$x \xleftrightarrow{FS} c_n$$

$$y \xleftrightarrow{FS} ?$$



Suppose $y \xleftrightarrow{FS} d_n$



$$d_n = \frac{1}{T}\int_0^T y(t).e^{-jn\omega t}dt$$

$$= \frac{1}{T}\int_0^T x(t-t_0).e^{-jn\omega t}dt$$

$$= \frac{1}{T}\int_0^T x(t-t_0).e^{-jn\omega(t-t_0+t_0)}.dt.$$

Change of variable : $t - t_0 = u$

$$d_n = \frac{1}{T}\int_{-t_0}^{T-t_0} x(u).e^{-jn\omega u}.e^{-jn\omega t_0}du$$

$$= e^{-jn\omega t_0}\frac{1}{T}\int_{-t_0}^{T-t_0} x(u).e^{-jn\omega u}du$$

$$d_n = c_n.e^{-jn\omega t_0}$$

3. <u>Frequency</u> <u>Shifting</u>.

$y(t) = x(t).e^{jk\omega t}$, k is some Integer

$$x(t) \xleftrightarrow{FS} c_n$$

$$y(t) \xleftrightarrow{FS} d_n$$

$$d_n = \frac{1}{T}\int_0^T x(t).e^{jk\omega t}.e^{-jn\omega t}dt$$

$$= \frac{1}{T}\int_0^T x(t).e^{-j(n-k)\omega t}dt.$$

$$= c_{n-k}.$$